



Aggre/Gator

event streams, swallowed whole

Intelligent Event Aggregation Platform

The Problem

Modern systems produce floods of related events - but they arrive fragmented, out of order, and from many sources.

Scattered Sources

Events arrive from APIs, queues, databases, webhooks - all in isolation with no shared context.

No End-to-End View

Individual events tell you something happened. You can't see the full story without manual correlation.

Lifecycle is Invisible

When did this order actually complete? Did this trade settle? Is this session still active?

Insights are Missing

Without grouped data, reporting is limited to raw event counts - not meaningful business metrics.

What Aggre/Gator Does

One platform. Any source. Complete event lifecycle - automatically.

01 Ingest from Anywhere

A single HTTP endpoint accepts events from any producer - APIs, queues, webhooks, batch jobs. Full REST API available for headless integration of all core features.

02 Group by Configurable Key

Policies define which field in the event body acts as the aggregation key. All events sharing a key value are automatically linked into one group.

03 Cradle-to-Grave Lifecycle

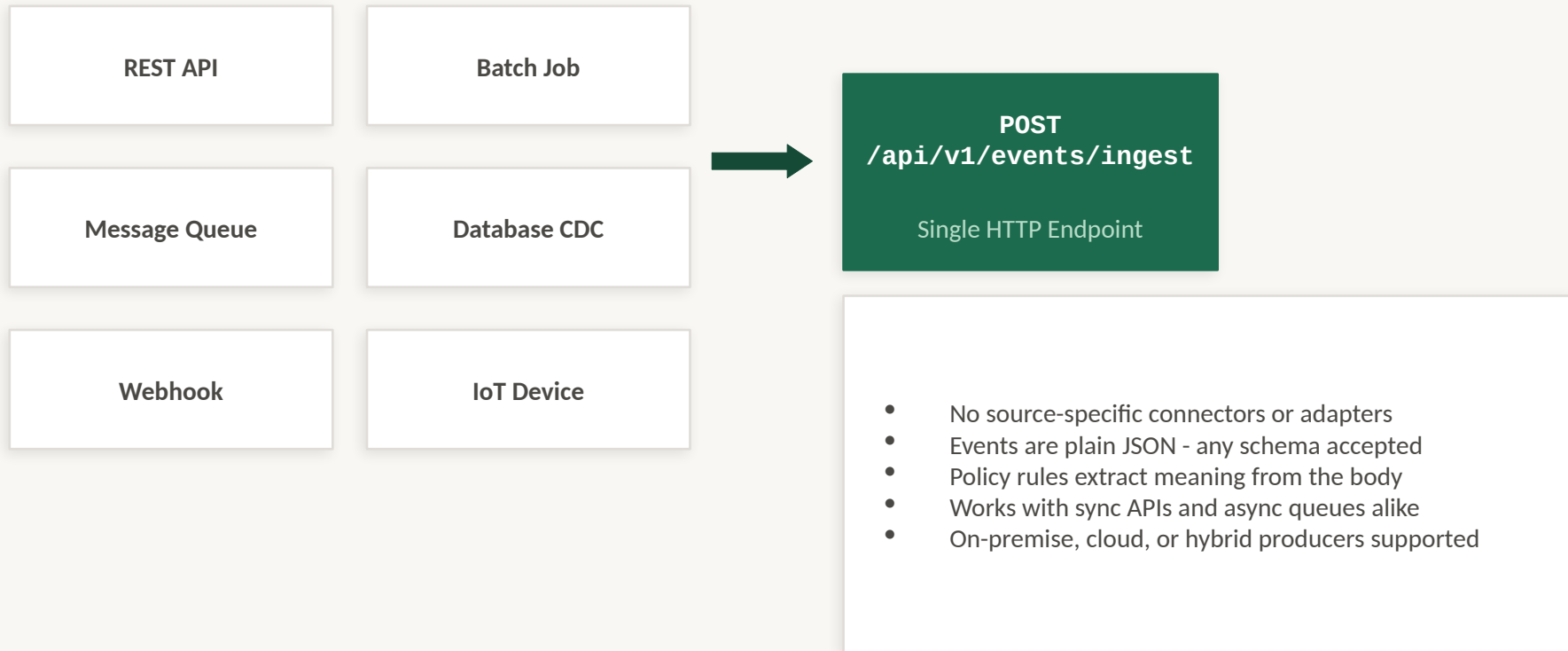
Define exactly which event opens a group (cradle) and which closes it (grave) using per-policy field conditions. You control the lifecycle rules.

04 Reports & Insights

Real-time dashboards show throughput, completion rates, duration distributions, and per-policy breakdowns. Timed-out groups are surfaced automatically.

Source Agnostic

One endpoint. Any producer.



Grouping by Configurable Key

Related events automatically linked - no code required.

Policy Configuration

```
name:      Trade Lifecycle
key_field: body.tradeRef
cradle_field: body.eventType
cradle_value: "trade.initiated"
grave_field: body.status
grave_value: "settled"
```



Event Group: TRD-9001



trade.initiated

CRADLE



trade.confirmed



trade.cleared



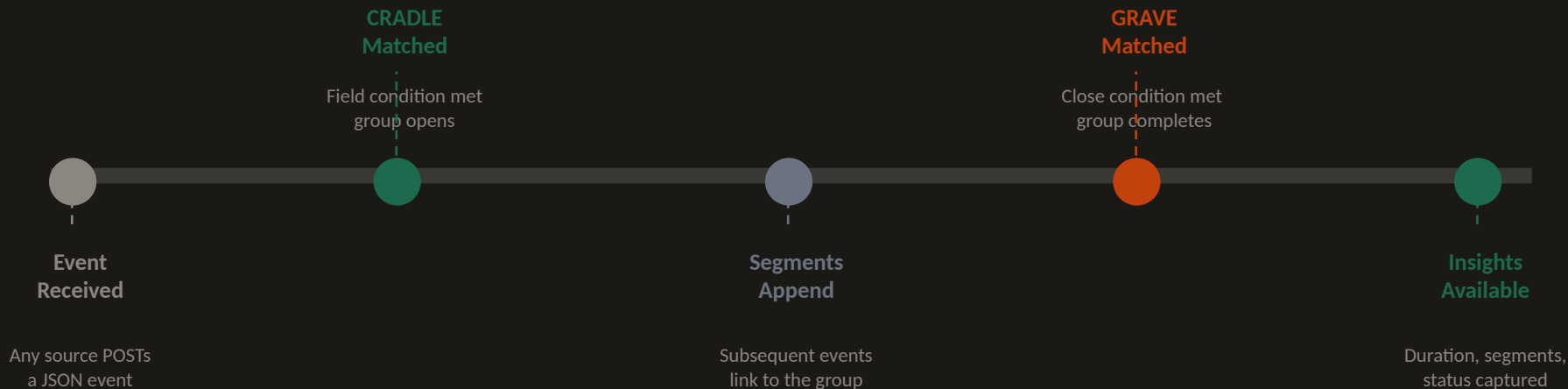
status: "settled"

GRAVE

Any JSON field can be the aggregation key — tradeRef, orderId, sessionId, correlationId. No schema changes required. Policies are created and modified through the UI at runtime.

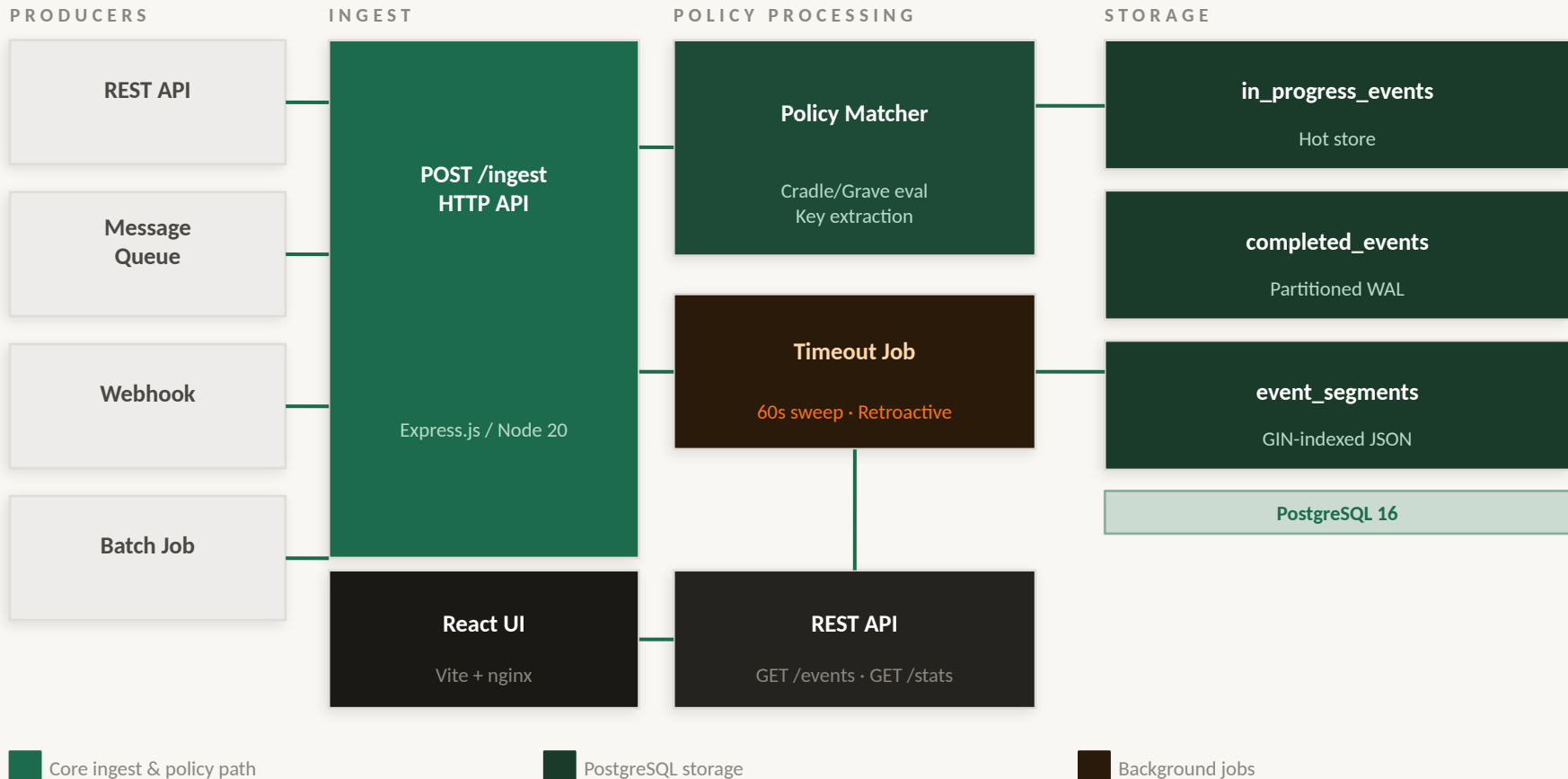
Cradle-to-Grave Lifecycle

User-defined open and close criteria. The platform handles the rest.



Auto-Timeout: if no grave arrives within a configurable window, the group is automatically closed and marked Timed Out — no manual intervention needed.

Architecture



REST API — Headless Integration

Every core feature is available programmatically - no UI required.

POST	<code>/api/v1/events/ingest</code>	Ingest an event from any source
GET	<code>/api/v1/events</code>	List & filter event groups (status, policy, date, key, body)
GET	<code>/api/v1/events/:id</code>	Retrieve a single group with all segments
GET	<code>/api/v1/events/stats</code>	Aggregate metrics — completion rate, throughput, counts
GET	<code>/api/v1/events/performance</code>	Slowest groups, in-progress aging, duration histogram
GET	<code>/api/v1/policies</code>	List all aggregation policies
POST	<code>/api/v1/policies</code>	Create a new policy
PUT	<code>/api/v1/policies/:id</code>	Update policy including timeout configuration
PATCH	<code>/api/v1/policies/:id/toggle</code>	Activate or deactivate a policy

All endpoints accept and return JSON · Authentication can be layered via reverse proxy or API gateway

Reports & Insights

From raw event streams to actionable intelligence - in real time.

91%

Completion Rate

3.2s

Avg Duration

0.8%

Timeout Rate

16,060

Total Segments

Throughput Charts

Group opens vs closes over time. Spot ingestion spikes and quiet periods at a glance.

Policy Breakdown Pies

Event groups and segment counts by policy — understand which flows dominate.

Hourly Activity Heatmap

7-day × 24-hour grid. Reveal peak windows and off-hours anomalies instantly.

Duration Histogram

See where groups cluster — sub-second vs multi-day. Identify slow outliers.

Policy Health Cards

Completion rate, timeout rate, avg/p50/p95 duration per policy with health badges.

Event Group Performance

Slowest groups, aging in-progress, and inline segment timeline drill-down.

Aggre/Gator

*event streams,
swallowed whole*

v1.0

In Summary

Source agnostic

One HTTP endpoint accepts events from any producer, any format, any system.

Configurable grouping

Policies define the key field that links related events - no code changes needed.

Cradle-to-grave lifecycle

User-defined open and close conditions give complete visibility into each group's journey.

Rich reporting

Real-time dashboards, per-policy health, duration analysis, and stale group detection.

Auto-timeout

Groups that never receive a grave are automatically aged out and surfaced in the UI.

Appendix

Screenshots from WiP Prototype

Current Status

WiP - Functional Prototype



Fully Dockerized Container

encapsulates the entire application, including its code, runtime, system tools, libraries, database and configurations, into a single, portable, and reproducible unit



Fully functional

All features are working as designed, no smoke and mirrors or hard coded data

Event Groups — Overview

Key Points

- Filter by status (Completed / In Progress / Timed Out), policy, date range, aggregation key and body content
- Stats bar shows live counts — total groups, completion rate, in-progress, timed-out, segments
- Rows per page and column visibility are configurable; export to JSON or CSV from any row
- Clicking a row opens the in-line detail panel without leaving the list

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event streams, swallowed whole

Auto refresh

Event Groups

Reports

Policies (4)

Ingest Event

1574

EVENT GROUPS

1455

COMPLETED

92% of total

119

IN PROGRESS

8% of total

0

TIMED OUT

4

POLICIES

5092

SEGMENTS

All statuses

All policies

Last 30d

KEY TSD-0001/aaa-1000...

BODY TradeRef-SMITH OF "t-smith"

Clear all

1574 total event group GET /api/v1/events?status=all&from=2026-04-18T23:00:00.000Zto=2026-05-11T22:59:59.999Zpage=1

Event Groups

Segments

Rows per page 25

Export

Columns

EVENT GROUP ID	POLICY	AGGREGATION KEY	SOURCE	STATUS	START TIME	SEG
c39c8b00-9a43-4cc9-b678-5ea0d...	EXAMPLE - API Request/Response	req-000328	-body.correlationId	Completed	08/05/2026, 23:19:09	2
739a02b-5a2b-4872-a06a-3229d...	EXAMPLE - Order Flow	ORD-010272	-body.orderId	Completed	08/05/2026, 22:50:36	6
7e6d63d-484a-4c84-56ea-c313f...	EXAMPLE - API Request/Response	req-000218	-body.correlationId	Completed	08/05/2026, 22:47:06	2
7e1f435-0313-4221-8baa-c4d6d...	EXAMPLE - API Request/Response	req-000667	-body.correlationId	Completed	08/05/2026, 22:42:07	2
60f9b4d-043d-4771-bb3b-b2c0c...	EXAMPLE - API Request/Response	req-000612	-body.correlationId	Completed	08/05/2026, 22:21:17	2
612c02d4-a43d-4485-86a0-2774e...	EXAMPLE - Trade Lifecycle	TRD-001299	-body.tradeRef	Completed	08/05/2026, 22:13:14	5
80d550b-0d0f-4f91-af0c-a467e...	EXAMPLE - Order Flow	ORD-010344	-body.orderId	Completed	08/05/2026, 22:02:46	6
4d02813c-1c0b-4c3b-bc79-1c0af...	EXAMPLE - API Request/Response	req-001057	-body.correlationId	Completed	08/05/2026, 21:46:02	2
439a0b0f-4ba1-439c-bc08-0bc0d...	EXAMPLE - API Request/Response	req-000381	-body.correlationId	Completed	08/05/2026, 20:55:37	2
1b000002-0000-0000-0000-00000...	EXAMPLE - Trade Lifecycle	TRD-9904	-body.tradeRef	In Progress	08/05/2026, 20:04:05	4
1b000001-0000-0000-0000-00000...	EXAMPLE - User Session	sess-10004-D	-body.sessionId	In Progress	08/05/2026, 20:00:05	4
3ba429f6-f11d-4700-a639-f6437...	EXAMPLE - API Request/Response	req-000283	-body.correlationId	Completed	08/05/2026, 19:58:03	2
cc000012-0000-0000-0000-00000...	EXAMPLE - API Request/Response	req-A004	-body.correlationId	Completed	08/05/2026, 19:52:05	2
cc000003-0000-0000-0000-00000...	EXAMPLE - User Session	sess-10003-C	-body.sessionId	Completed	08/05/2026, 19:27:05	2
cc000021-0000-0000-0000-00000...	EXAMPLE - API Request/Response	req-A003	-body.correlationId	Completed	08/05/2026, 19:17:05	2
cc000006-0000-0000-0000-00000...	EXAMPLE - Trade Lifecycle	TRD-9903	-body.tradeRef	Completed	08/05/2026, 19:02:05	3
cc000010-0000-0000-0000-00000...	EXAMPLE - API Request/Response	req-A002	-body.correlationId	Completed	08/05/2026, 18:27:05	2
1b000003-0000-0000-0000-00000...	EXAMPLE - Order Flow	ORD-10047	-body.orderId	In Progress	08/05/2026, 18:12:05	4
cc000009-0000-0000-0000-00000...	EXAMPLE - API Request/Response	req-A001	-body.correlationId	Completed	08/05/2026, 17:42:05	2
79ad3203-a35d-42a9-a0d2-44970...	EXAMPLE - API Request/Response	req-000665	-body.correlationId	Completed	08/05/2026, 17:29:09	2
6070402-d318-444c-90ab-90ab0...	EXAMPLE - API Request/Response	req-000999	-body.correlationId	In Progress	08/05/2026, 17:20:44	2
cc000005-0000-0000-0000-00000...	EXAMPLE - Trade Lifecycle	TRD-9902	-body.tradeRef	Completed	08/05/2026, 16:42:05	3
cc000002-0000-0000-0000-00000...	EXAMPLE - User Session	sess-10002-B	-body.sessionId	Completed	08/05/2026, 15:57:05	5
cc000004-0000-0000-0000-00000...	EXAMPLE - Trade Lifecycle	TRD-9901	-body.tradeRef	Completed	08/05/2026, 15:12:05	4
b3f9f07d-a06a-40ff-0522-c731d...	EXAMPLE - Order Flow	ORD-010526	-body.orderId	Completed	08/05/2026, 14:49:51	5

Prev

1

2

63

Next

Page 1 of 61

completed_events — double (N/A)

In progress_events — hot

Policy-driven — key, trade & give from body

Key path resolved from segment body

Fig. A1 — Event Groups — Overview

Event Group Detail & Timeline

Key Points

- Left panel shows the full event group — aggregation key, policy, start/end, duration and store
- Each segment is listed in sequence; CRADLE and GRAVE are labelled; LAST BEFORE TIMEOUT marks auto-closed groups
- Click any segment to expand the raw JSON body for inspection
- Switch between List and Timeline views; export the full group as JSON or CSV

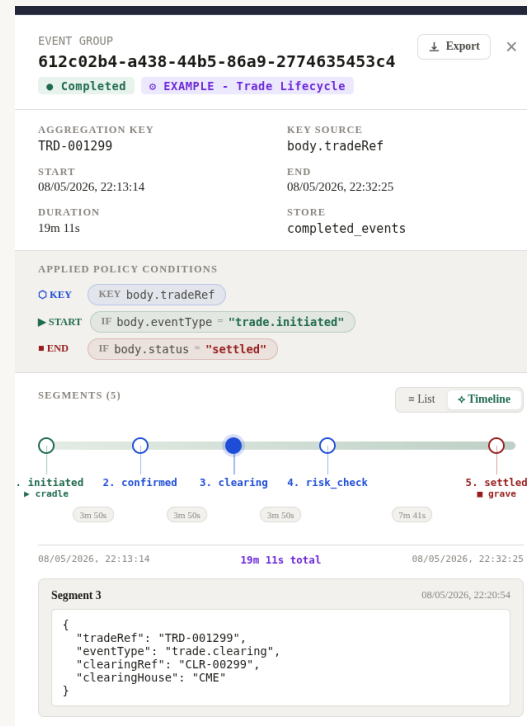
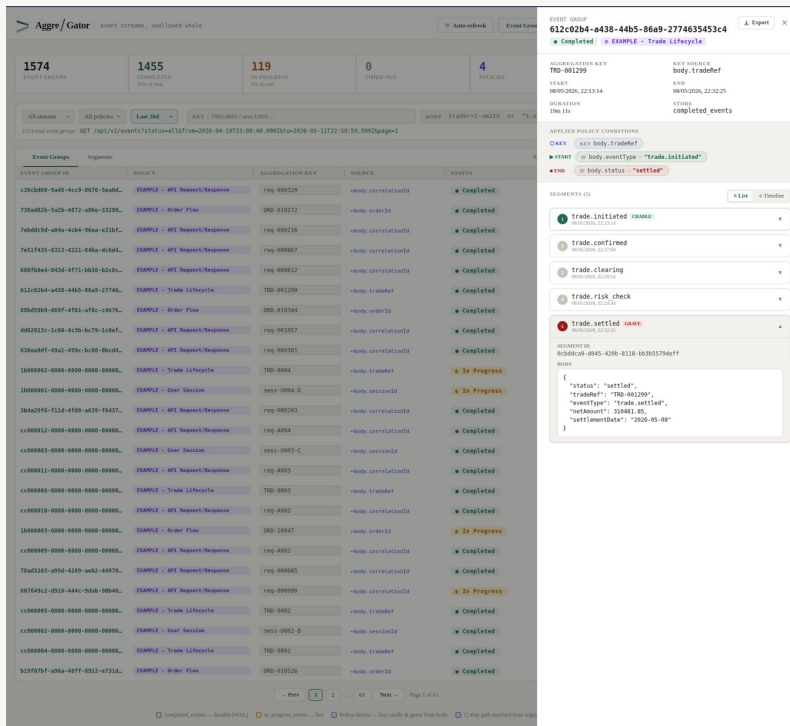


Fig. A2 — Event Group Detail & Timeline

Raw Events

A3

Key Points

- The Segments tab shows every individual event across all groups in a flat, sortable list
- Filter segments by policy, date range, aggregation key or body search
- Useful for finding a specific event ID and jumping directly to its parent group
- Column visibility is configurable — show or hide event ID, timestamp, body fields as needed

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Auto-refresh

Event Groups

Reports

Policies (4)

Import Event

5015

EVENT GROUPS

4694

COMPLETED

96% of total

321

IN PROGRESS

0% of total

1

TIMED OUT

0% of total

4

POLICIES

16172

SEGMENTS

All statuses ▾

All policies ▾

Last 24h ▾

KEY TRD-0001 / sess.U001...

BODY traderet-smith or "t-smith" ▾

No filters

5015 total event groups GET /api/v1/events?status=all&page=1

Event Groups		Segments		Rows per page: 25 ▾ A. Export Columns									
SEGMENT ID	EVENT GROUP ID	POLICY	AGGREGATION KEY	SEQ	EVENT TYPE	CRABLE	GRAVE	RECEIVED AT					
f519566-6569-6efe-b14b...	739a092b-5a2b-4872-ab8e...	EXAMPLE - Order Flow	ORD-010272	6	order.delivered	—	GRAVE	11/05/2026, 09:58:58					
2f57796b-ee09-4714-9676...	739a092b-5a2b-4872-ab8e...	EXAMPLE - Order Flow	ORD-010272	5	order.out for delivery	—	—	11/05/2026, 21:36:43					
ce6ed6bc-da75-6daa-91e1...	88d5909-869f-4781-af9c...	EXAMPLE - Order Flow	ORD-010344	6	order.delivered	—	GRAVE	11/05/2026, 14:11:44					
57769794-c5c5-42b0-ae9f...	88d5909-869f-4781-af9c...	EXAMPLE - Order Flow	ORD-010344	5	order.out for delivery	—	—	11/05/2026, 04:34:23					
fe1e7204-91d7-4269-aa2b...	b19f67bf-a96a-48ff-8922...	EXAMPLE - Order Flow	ORD-010526	5	order.delivered	—	GRAVE	11/05/2026, 00:54:41					
eddc181e-6e03-4f1e-b7a9...	739a092b-5a2b-4872-ab8e...	EXAMPLE - Order Flow	ORD-010272	4	order.shipped	—	—	10/05/2026, 16:24:47					
25f7c7b4-4246-4990-b6de...	88d5909-869f-4781-af9c...	EXAMPLE - Order Flow	ORD-010344	4	order.shipped	—	—	10/05/2026, 06:07:15					
ed8e408b-3dc5-49f4-813c...	739a092b-5a2b-4872-ab8e...	EXAMPLE - Order Flow	ORD-010272	3	order.picked	—	—	09/05/2026, 23:47:07					
95ca79ab-6899-4d17-871b...	b19f67bf-a96a-48ff-8922...	EXAMPLE - Order Flow	ORD-010526	4	order.shipped	—	—	09/05/2026, 19:52:16					
12f5bfe2-4102-4d47-b7a5...	88d5909-869f-4781-af9c...	EXAMPLE - Order Flow	ORD-010344	3	order.picked	—	—	09/05/2026, 17:17:28					
e373e9c-f214-43ab-bdae...	b19f67bf-a96a-48ff-8922...	EXAMPLE - Order Flow	ORD-010526	3	order.picked	—	—	09/05/2026, 08:15:18					
18313c0b-6930-4f00-ab1d...	739a092b-5a2b-4872-ab8e...	EXAMPLE - Order Flow	ORD-010272	2	order.payment	—	—	09/05/2026, 00:30:22					
1f523813-689e-4d8f-ab9d...	88d5909-869f-4781-af9c...	EXAMPLE - Order Flow	ORD-010344	2	order.payment	—	—	09/05/2026, 23:19:45					
b470b390-565f-4297-ba27...	c38cb09-9a45-4cc9-9a7e...	EXAMPLE - API Request/Response	req-000320	2	api.response	—	GRAVE	09/05/2026, 23:19:12					
55daec2b-fd41-450b-bd6e...	c38cb09-9a45-4cc9-9a7e...	EXAMPLE - API Request/Response	req-000320	1	api.request	CRABLE	—	09/05/2026, 23:19:09					
d6662073-2602-4116-bac4...	739a092b-5a2b-4872-ab8e...	EXAMPLE - Order Flow	ORD-010272	1	order.created	CRABLE	—	09/05/2026, 22:56:36					
892b7136-1454-4bd7-9910...	7ab86c9d-a96a-4c34-98ea...	EXAMPLE - API Request/Response	req-000216	2	api.response	—	GRAVE	09/05/2026, 22:47:06					
55608429-8c9e-4443-ad78...	7ab86c9d-a96a-4c34-98ea...	EXAMPLE - API Request/Response	req-000216	1	api.request	CRABLE	—	09/05/2026, 22:47:06					
3d68adda-094d-4632-4346...	7a51f435-8313-4221-84ba...	EXAMPLE - API Request/Response	req-000867	2	api.response	—	GRAVE	09/05/2026, 22:42:07					
f9a19040-429d-4c81-a697...	7a51f435-8313-4221-84ba...	EXAMPLE - API Request/Response	req-000867	1	api.request	CRABLE	—	09/05/2026, 22:42:07					
8cb8dca8-0045-4280-8118...	612c8204-0438-4a85-88a9...	EXAMPLE - Trade Lifecycle	TRD-001299	5	trade.settled	—	GRAVE	09/05/2026, 22:32:25					
789d2c0e-da75-46d6-939d...	612c8204-0438-4a85-88a9...	EXAMPLE - Trade Lifecycle	TRD-001299	4	trade.risk check	—	—	09/05/2026, 22:34:41					
994d991-05f8-48bd-9e00...	68879e4-843d-4771-3b39...	EXAMPLE - API Request/Response	req-000612	2	api.response	—	GRAVE	09/05/2026, 22:21:17					
883ecff2-8a2a-4a84-b448...	68879e4-843d-4771-3b39...	EXAMPLE - API Request/Response	req-000612	1	api.request	CRABLE	—	09/05/2026, 22:21:17					
11111111-1111-1111-1111...	61111111-1111-1111-1111...	EXAMPLE - Trade Lifecycle	TRD-001299	4	trade.settled	—	—	09/05/2026, 22:21:17					

Fig. A2 — Event Group Detail & Timeline

Raw Events – Drill into Event Group Detail

Key Points

- Clicking any segment row opens the parent event group detail panel on the right
- The originating segment is highlighted so context is immediately clear
- The full segment timeline is visible — navigate forward or back through related segments
- Group metadata (policy, duration, status) is shown alongside the segment body

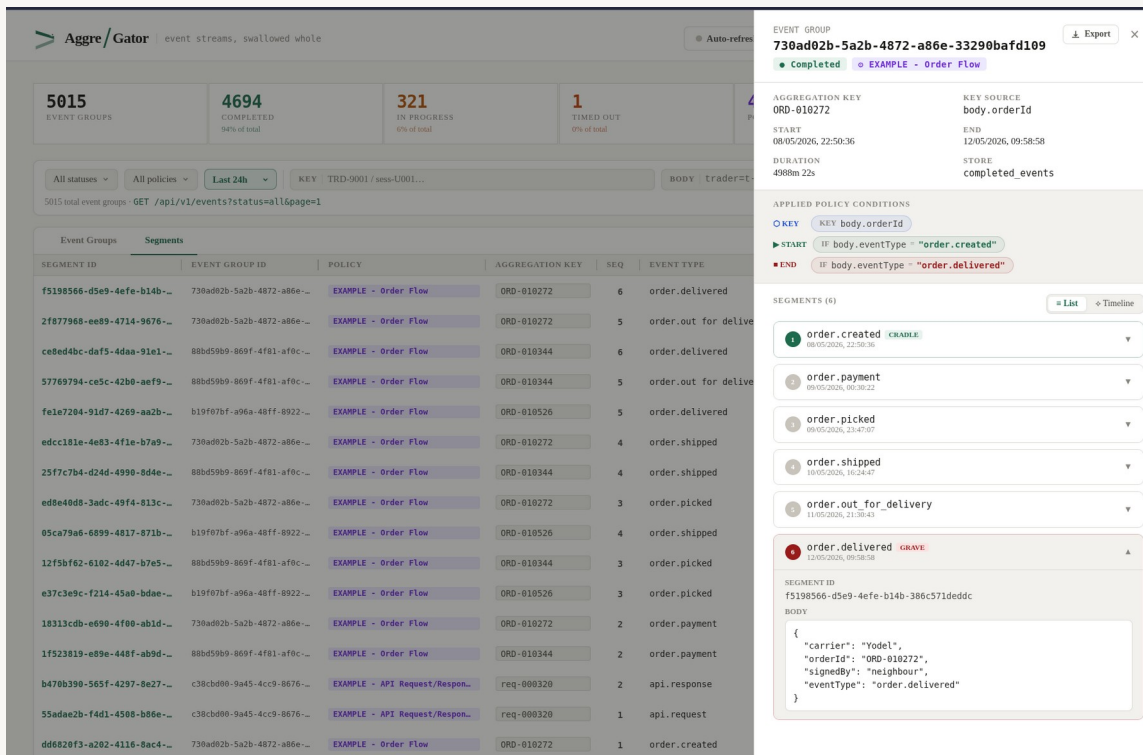


Fig. A2 — Event Group Detail & Timeline

Reports — Overview

Key Points

- Throughput charts show group opens vs closes over time with configurable bucket granularity
- Pie charts break down event groups and segment counts by policy for the selected time window
- Hourly heat map reveals peak ingestion windows across a 7-day × 24-hour grid
- KPI row below shows completion rate, avg segments per group, throughput and in-progress count

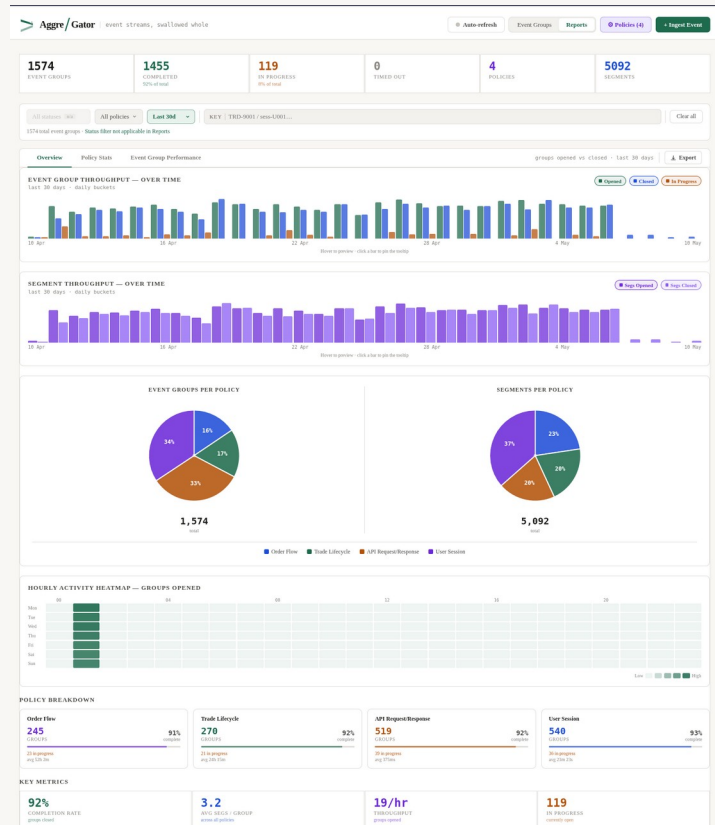


Fig. A3 — Reports — Overview

Reports — Policy Stats

Key Points

- Each policy gets a health card — Healthy / Warning / Critical based on completion rate
- Stat grid shows total, completed, timed-out, in-progress, segments and avg duration per policy
- Gradient completion bar shows natural completions vs timeouts in distinct colours
- Expand any card to see avg / p50 / p95 duration estimates with a relative scale bar

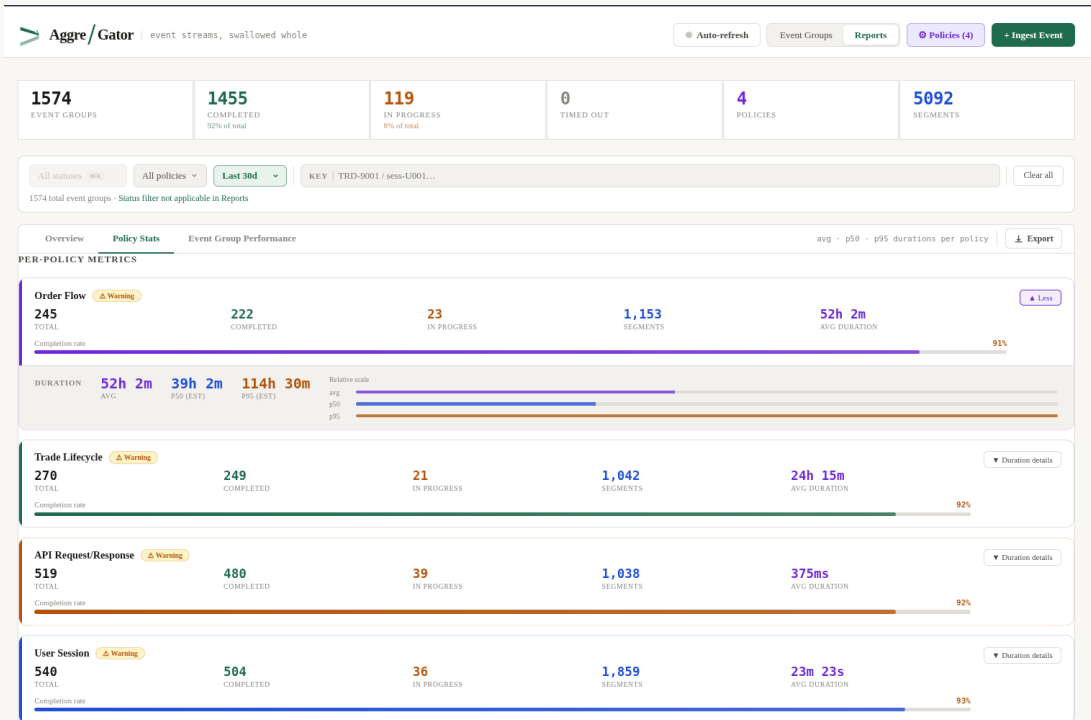


Fig. A4 — Reports — Policy Stats

- Summary stats bar shows slowest group, fastest completion, most segments, and stale-open count
- Slowest Completed table is sorted by duration — click Timeline to drill into any group inline
- In-Progress Aging table sorts oldest-first and flags groups open longer than 1 hour
- Duration histogram buckets all completed groups from sub-second to multi-day at a glance

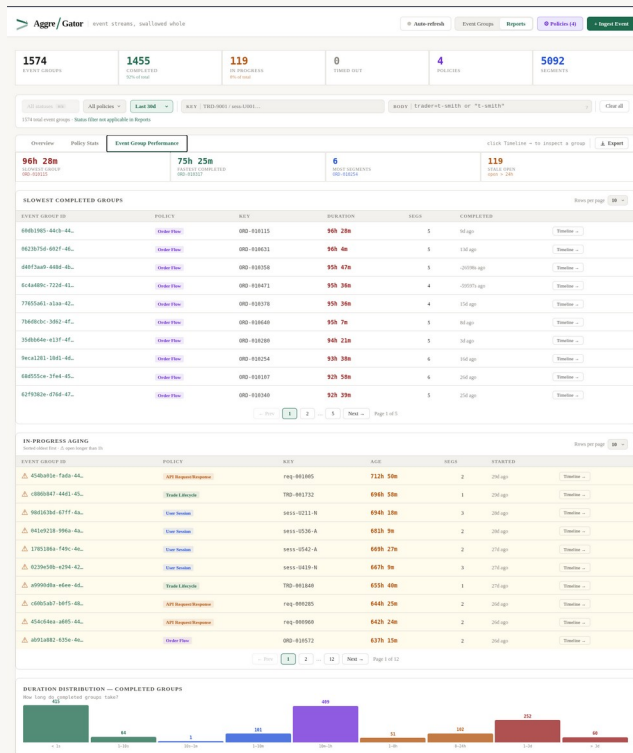


Fig. A5 — Event Group Performance

Policy Configuration

Key Points

- Each policy defines a key field (the grouping key), cradle condition (open) and grave condition (close)
- Auto-Timeout toggle sets a per-policy window — groups with no grave after this period are auto-closed
- Saving a timeout retroactively applies it: existing in-progress groups are swept and closed immediately
- Policies can be activated/deactivated at runtime without restarting — changes take effect on the next ingest

Aggregation Policies
4 active • 0 inactive

Search by name, domain, or description...

Active (4) Inactive (0)

EXAMPLE - User Session user * Active Edit
Aggregates user session events. Key extracted from body.sessionId
92f9b32a-ca85-4bee-83e9-181e64401a64 copy

- KEY body.sessionId → aggregation key
- body.eventType = "user.login" → open group
- body.eventType = "user.logout" → close group

Validate event body

EXAMPLE - Trade Lifecycle trade * Active Edit
Aggregates trade events. Key from body.tradeId; closes when body.status = settled
f157e596-916c-4d23-8917-9a83c8c8b04 copy

- KEY body.tradeId → aggregation key
- body.eventType = "trade.initiated" → open group
- body.status = "settled" → close group

Validate event body

EXAMPLE - API Request/Response api * Active Edit
Pairs API requests with responses. Key extracted from body.requestId
1ceef779-c634-48de-9108-84367a276c4 copy

- KEY body.requestId → aggregation key
- body.eventType = "api.request" → open group
- body.eventType = "api.response" → close group

Validate event body

EXAMPLE - Order Flow order * Active Edit
Aggregates order lifecycle. Key extracted from body.orderId
c32ef7f5-f64d-4ac4-8662-6302a30843c6 copy

- KEY body.orderId → aggregation key
- body.eventType = "order.created" → open group
- body.eventType = "order.delivered" → close group

Validate event body

Edit - EXAMPLE - User Session

Define correlation key, cradle condition, and grave condition

IDENTITY
POLICY NAME * EXAMPLE - User Session EVENT DOMAIN user *

DESCRIPTION
Aggregates user session events. Key extracted from body.sessionId

CONDITIONS — ALL THREE EVALUATED AGAINST EVERY INGESTED SEGMENT BODY

AGGREGATION KEY	CRADLE / START	GRAVE / END
FIELD PATH * sessionId Value at this path becomes the correlation key	FIELD PATH * eventType MUST EQUAL * user.login body.eventType new "user.login"	FIELD PATH * eventType MUST EQUAL * user.logout body.eventType new "user.logout"

Paths use dot-notation resolved against the full event body JSON. No lifecycle flags needed from the producer.

AUTO-TIMEOUT
Automatically close groups that stop receiving segments. Defaults to off — grave segment required to close.

☒ Close group after 24 hours since last segment received

Click starts at cradle and ends on each new segment. Group is promoted to 15min_off when no segment arrives within this window.

Delete Policy Cancel Save Changes

Fig. A6 — Policy Configuration

Self Hosted API Documentation

A8

Key Points

- Self hosted swagger API documentation at `<Sever>:<port>/api/v1/docs`
- Contains documentation and schema for all endpoints
- Open API 3.0 compliant JSON exportable (works in postman etc)
-

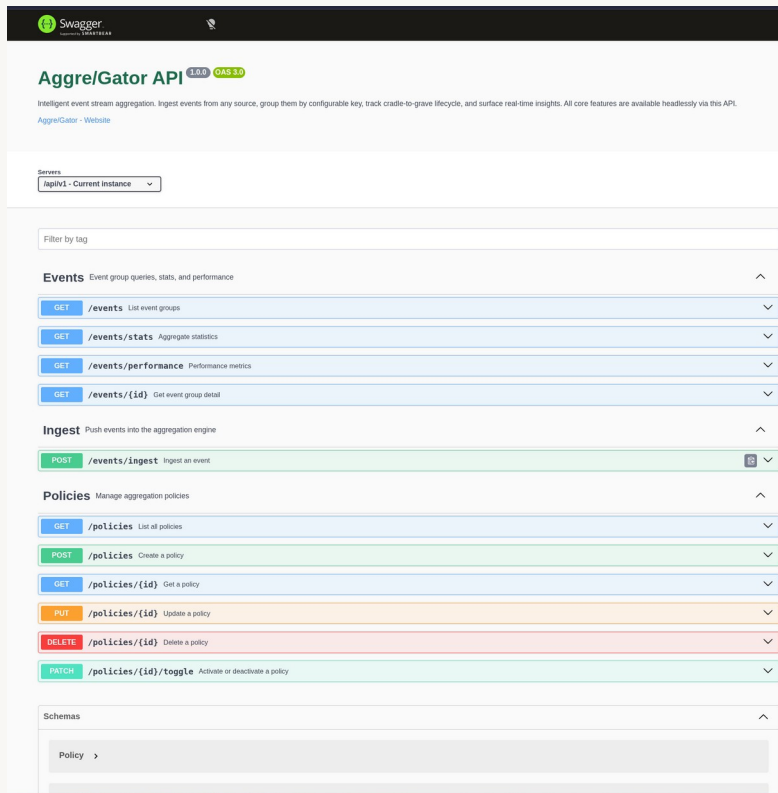


Fig. A6 — Policy Configuration

Wip Configuration Menus

Key Points

- “Burger” style menu item added for de-cluttering of UI / ease of configuration
- Policies screen migrated to Menu
- Accounts menu item and UI added
-

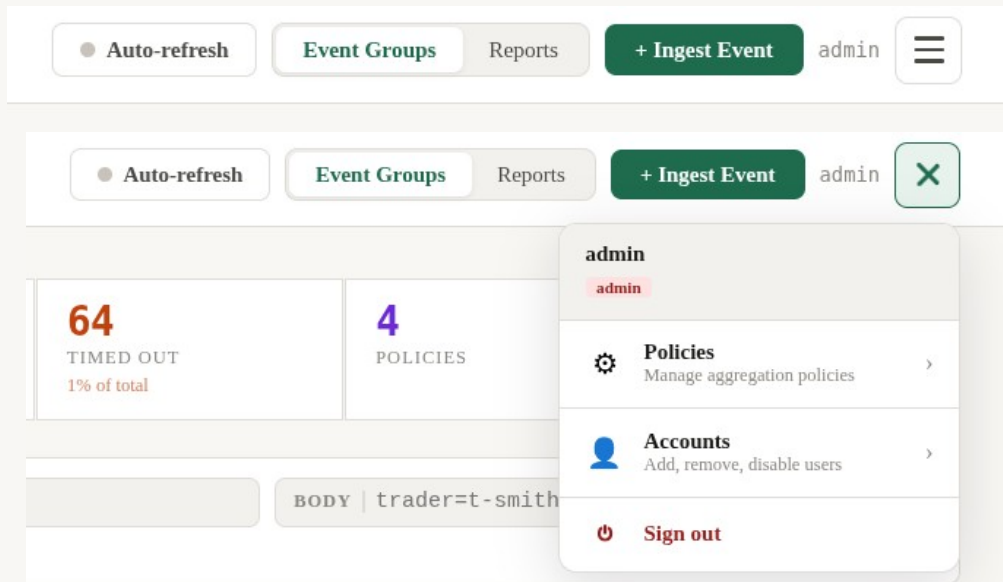


Fig. A6 — Policy Configuration

Wip Configuration Menus - Users

Key Points

- Manage UI accounts (Add,Disable,Remove)
- Grant Roles (View,Edit,FullAdmin) - *API Read/API Write in Futures
- Last Login host and datetime information
- All accounts stored in database with password hashing

The screenshot displays the 'Accounts' management interface. At the top, there are tabs for 'Event Groups', 'Reports', and '+ Ingest Event', along with a user profile 'admin' and a close button. The main section is titled '< Accounts' and includes a '+ Add' button. Below this, a list of accounts is shown, with the 'admin' account highlighted. The 'admin' account details include a profile icon 'A', the name 'admin', email 'admin@localhost', and last login '13/05/2026, 10:45'. A 'Disable' button is next to the account. Below the account list, a legend states: 'viewer = read only · editor = ingest + policies · admin = full access'.

The bottom part of the screenshot shows the 'New account' form. It has a 'Cancel' button at the top right. The form fields are: 'username', 'email', and 'password'. Below these fields, the 'Role' is set to 'viewer', with 'editor' and 'admin' as options. A 'Create account' button is at the bottom of the form. Below the form, the 'admin' account is listed again with its details and a 'Disable' button.

Fig. A6 — Policy Configuration